

PGCP-FBD-2026

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Course Name: Devops and Challenges in Fintech & Blockchain

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Ques1. What is the command to initialize a new Git repository?

Ans - git init

```
PS C:\Users\alxgr\Desktop> mkdir GitAssignment

Directory: C:\Users\alxgr\Desktop

Mode                LastWriteTime         Length Name
----                -
d-----          20-05-2026         17:23         GitAssignment

PS C:\Users\alxgr\Desktop> cd .\GitAssignment\
PS C:\Users\alxgr\Desktop\GitAssignment> git init
Initialized empty Git repository in C:/Users/alxgr/Desktop/GitAssignment/.git/
PS C:\Users\alxgr\Desktop\GitAssignment> |
```

Ques2. What is the command to add files to the staging area?

Ans - git add <filename>

```
PS C:\Users\alxgr\Desktop\GitAssignment> echo Hello World > test.txt
PS C:\Users\alxgr\Desktop\GitAssignment> git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    test.txt

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\alxgr\Desktop\GitAssignment> git add .\test.txt
PS C:\Users\alxgr\Desktop\GitAssignment> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   test.txt

PS C:\Users\alxgr\Desktop\GitAssignment>
```

Ques3. What is the command to commit changes with a message?

Ans - `git commit -m "<message>"`

```
PS C:\Users\alxgr\Desktop\GitAssignment> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   test.txt

PS C:\Users\alxgr\Desktop\GitAssignment> git commit -m "First Commit"
[master (root-commit) d5a21b5] First Commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 test.txt
PS C:\Users\alxgr\Desktop\GitAssignment> git status
On branch master
nothing to commit, working tree clean
PS C:\Users\alxgr\Desktop\GitAssignment> |
```

Ques4. What is the command to check the status of the repository?

Ans - `git status`

```
PS C:\Users\alxgr\Desktop\GitAssignment> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   test.txt

PS C:\Users\alxgr\Desktop\GitAssignment> git commit -m "First Commit"
[master (root-commit) d5a21b5] First Commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 test.txt
PS C:\Users\alxgr\Desktop\GitAssignment> git status
On branch master
nothing to commit, working tree clean
PS C:\Users\alxgr\Desktop\GitAssignment> |
```

Ques5. What is the command to create a new branch and switch to it?

Ans - `git checkout -b "<branch_name>"`

```
PS C:\Users\alxgr\Desktop\GitAssignment> git branch
* master
PS C:\Users\alxgr\Desktop\GitAssignment> git checkout -b "Temp_Branch"
Switched to a new branch 'Temp_Branch'
PS C:\Users\alxgr\Desktop\GitAssignment> git branch
* Temp_Branch
  master
PS C:\Users\alxgr\Desktop\GitAssignment> |
```

Ques6. What is the command to merge a branch into the current branch?

Ans - git merge <branch_name>

```
PS C:\Users\alxgr\Desktop\GitAssignment> git merge Temp_Branch
Updating d5a21b5..0aac860
Fast-forward
 test2.txt | Bin 0 -> 38 bytes
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 test2.txt
PS C:\Users\alxgr\Desktop\GitAssignment> dir
```

Directory: C:\Users\alxgr\Desktop\GitAssignment

Mode	LastWriteTime	Length	Name
-a----	20-05-2026 17:29	30	test.txt
-a----	20-05-2026 17:42	38	test2.txt

```
PS C:\Users\alxgr\Desktop\GitAssignment> |
```

Ques7. What is the command to view the differences between two branches or commits?

Ans - git diff <branch1> <branch2>

git diff <commit1> <commit2>

```
PS C:\Users\alxgr\Desktop\GitAssignment> git diff Temp_Branch master
diff --git a/test2.txt b/test2.txt
index b9332c2..1553a29 100644
Binary files a/test2.txt and b/test2.txt differ
PS C:\Users\alxgr\Desktop\GitAssignment> |
```

```
PS C:\Users\alxgr\Desktop\GitAssignment> git diff 6ddb0a05e864d309678f21e6a079f6c189fde33 d5a21b5d76195794d66fb1127fb7aa978f32c5d8
diff --git a/test2.txt b/test2.txt
deleted file mode 100644
index b9332c2..0000000
Binary files a/test2.txt and /dev/null differ
PS C:\Users\alxgr\Desktop\GitAssignment> ^S|
```

Ques8. What is the command to revert changes to a specific file?

Ans - git restore --source <commit> <file>

```
PS C:\Users\alxgr\Desktop\GitAssignment> git restore --source cae46dd1fac81de776b60aa4c1524e858ab334f2 test2.txt
PS C:\Users\alxgr\Desktop\GitAssignment> |
```

Ques9. What is the command to undo the last commit?

Ans -

To undo the commit but keep your changes staged/unstaged: **git reset --soft HEAD~1**

To completely nuclear-delete the last commit and all its changes: **git reset --hard HEAD~1**

```
PS C:\Users\alxgr\Desktop\GitAssignment> git reset --hard HEAD~1
HEAD is now at cae46dd Revert "Adding changes"
PS C:\Users\alxgr\Desktop\GitAssignment> git log
commit cae46dd1fac81de776b60aa4c1524e858ab334f2 (HEAD -> Temp_Branch)
Author: AlexAshwin <alxgrg123@gmail.com>
Date:   Wed May 20 22:15:19 2026 +0530

    Revert "Adding changes"

    This reverts commit a2b4c88f90f6682464728d73d9c9e2f7f6507dfd.

commit a2b4c88f90f6682464728d73d9c9e2f7f6507dfd
Author: AlexAshwin <alxgrg123@gmail.com>
Date:   Wed May 20 22:03:17 2026 +0530

    Adding changes

commit cc14cab5dba415e8a181e3d36e36c4b3767ff757
Author: AlexAshwin <alxgrg123@gmail.com>
Date:   Wed May 20 18:09:52 2026 +0530

    Third commit edit test2 file

commit 6ddbd0a05e864d309678f21e6a079f6c189fde33
Author: AlexAshwin <alxgrg123@gmail.com>
Date:   Wed May 20 17:45:17 2026 +0530

    Third commit editing test2.txt
```

Ques10. What is the command to push changes to a remote repository?

Ans - git push origin <branch_name>

```
PS C:\Users\alxgr\Desktop\GitAssignment> git remote add origin https://github.com/AlexAshwin/temp.git
PS C:\Users\alxgr\Desktop\GitAssignment> git branch -M main
PS C:\Users\alxgr\Desktop\GitAssignment> git push -u origin main
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 12 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (17/17), 1.57 KiB | 402.00 KiB/s, done.
Total 17 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/AlexAshwin/temp.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```